

Hui Lin

Machine Learning Engineer · Computer Vision · Scalable Systems

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Professional Summary

Machine Learning Engineer specializing in **Deep Learning Architectures, Computer Vision, and Time-Series Analysis**. Proven track record of translating complex domain-specific workflows into scalable AI solutions for highly constrained physiological signals and high-resolution imaging. Expert in resolving long-tail data imbalances and designing robust validation strategies for production-grade models. Ph.D. graduate (Sept 2025) with 13 first-author papers, **1,153+ citations**, and multiple **Top-5 global algorithm rankings** (MICCAI/ISBI).

Skills

ML Architectures: Transformers, Self-Attention, YOLOv8, U-Net, GANs, GNNs, Contrastive Learning, UDA

Infrastructure & Tools: PyTorch, C++, Python, CUDA, SQL (Spark/Hive), MLOps, Docker, Git, Linux

Core Domains: Computer Vision, Representation Learning, Large-Scale Content Recommendation

Global Algorithm Competitions

MICCAI 2024: MyoPS++ (2nd) & FLARE (5th) **ISBI 2024:** JustRAIGS (5th) **MICCAI 2023:** ARCADE (3rd)

Selected Industry and Research Experience

Representation Learning & Scalable MLOps Pipelines 06.2024 – 08.2024, 02.2025 – Present
OPPO US Research Center Palo Alto, CA

- Led the production deployment of scalable **MLOps batch pipelines** (Spark/Hive) for a content recommendation engine serving **8M+ users**, executing rigorous A/B testing protocols to optimize ranking algorithms.
- Architected highly efficient **Transformer** and ResNet models for 800K+ sequential signals (PPG/ECG), analyzing continuous streams to extract predictive metrics from highly constrained hardware (earned the **2025 Polaris Team Award**).
- Engineered validation strategies using **contrastive learning** to map noisy, unstructured sensor data to stationary clinical ground-truths, ensuring high-fidelity monitoring in free-living environments.
- Developed a multi-task regression model predicting temporal changes via feature subtraction of paired sequential inputs.

Computer Vision & Deep Learning Architectures 03.2020 – 09.2025
Northwestern University Evanston, IL

- Engineered a self-attention vision model for large-scale clinical screening, integrating a **YOLOv8** localization pipeline that boosted diagnostic sensitivity to **85.70%** (at 95% specificity) and secured **5th place globally** (ISBI JustRAIGS).
- Built **Unsupervised Domain Adaptation (UDA)** frameworks using GANs to improve model robustness across heterogeneous datasets, achieving a **+11.4% DSC improvement** (**MICCAI FLARE 5th Place Globally**).
- Developed a **YOLO-to-UNet** ensemble optimized for pixel-wise object localization and fine-grained feature detection in high-resolution spatial imaging (**MICCAI ARCADE 3rd Place Globally**).
- Implemented a hybrid **GNN-GRU** architecture to forecast long-term physical states, validating predictive representations against complex simulations and ground-truth physics.

Anomaly Detection & Generative Data Strategy 2015 – 2019
Huazhong University of Science and Technology Wuhan, China

- Synthesized realistic high-resolution visual data via **CycleGAN** to resolve severe class imbalances in industrial datasets, optimizing model robustness and improving downstream detection accuracy by **2.95% (370 citations)**.
- Architected a deep **CNN with CAM** (Class Activation Mapping) for joint classification and fine-grained visual localization, achieving a low **5.04% error rate (303 citations)**.

Education

Ph.D. Electrical Engineering (GPA 3.9/4.0) 09.2019 – 09.2025
Northwestern University (Advised by A. Katsaggelos & D. Kim) Evanston, IL

M.S. Mechanical Engineering (Rank 1) 09.2016 – 06.2019
Huazhong University of Science and Technology Wuhan, China

B.S. Materials Processing (Rank 3, Qiming College) 09.2012 – 06.2016
Huazhong University of Science and Technology Wuhan, China

Selected Publications (13 First-Author Papers, 1,153+ Citations)

- Longitudinal Wrist PPG Analysis for Hypertension Risk Screening** | *ICASSP 2025* | Lin, H., et al.
- DRL-STNet: UDA for Cross-modality Medical Image Segmentation** | *MICCAI 2024* | Lin, H., et al.
- Defect Image Sample Generation with GAN...** | *IEEE TASE* | **370 citations**
- Automated Defect Inspection of LED Chip using Deep CNN** | *JIM* | **303 citations**